PHY 130A
Lecture Notes:
Fermi's Golden Rule

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1 The Interaction Picture

We consider a time-dependent Hamiltonian of the form:

$$H = H^{(0)} + \delta H(t)$$

where $H^{(0)}$ is time independent and so well understood, with energy eigenstates:

$$H^{(0)} |n\rangle = E_n |n\rangle$$

and the $\delta H(t)$ represents a new interaction. The new interaction is time dependent and therefore has no energy eigenstates.

The interaction picture factors out the simple and well-understood time dependence from $H^{(0)}$ from the to-be-determined time dependence of $\delta H(t)$ as:

$$|\Psi(t)\rangle = \sum_m c_m(t) \exp\left(-i\frac{E_m t}{\hbar}\right) |m\rangle$$

The $\{|m\rangle\}$ are still a basis for the full Hamiltonian H, even though they are not eigenstates, so this is certainly a valid expansion, due to the time dependent coefficients $c_m(t)$. This just says that $|\Psi(t)\rangle$ can be expressed at any time as a sum of basis vectors, which is certainly true. We will know it is useful if it simplifies the problem. Specifically, if the time dependence in the coefficients is due exclusively to $\delta H(t)$.

Applying the SE:

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t} |\psi(t)\rangle &= H\, |\psi(t)\rangle \\
&= H^{(0)}\, |\psi(t)\rangle + \delta H(t)\, |\psi(t)\rangle \\
\left(i\hbar\frac{\partial}{\partial t} - H^{(0)}\right) |\psi(t)\rangle &= \delta H(t)\, |\psi(t)\rangle
\end{aligned}$$

where we have arranged things in the hope of getting some cancellation on the left hand side:

$$\begin{aligned}
\left(i\hbar\frac{\partial}{\partial t} - H^{(0)}\right) |\psi(t)\rangle &= \sum_m \{i\hbar\dot{c}_m(t) + E_m c_m(t) - E_m c_m(t)\} \exp\left(-i\frac{E_m t}{\hbar}\right) |m\rangle \\
&= \sum_m i\hbar\dot{c}_m(t)\, \exp\left(-i\frac{E_m t}{\hbar}\right) |m\rangle
\end{aligned}$$

and indeed we do. Meanwhile the RHS is:

$$\delta H(t)\,|\psi(t)\rangle = \sum_m c_m(t)\,\exp\left(-i\frac{E_m t}{\hbar}\right)(\delta H(t)\,|m\rangle)$$

and so we cannot match coefficients term by term in the expansion because:

$$|m\rangle \;\neq\; \delta H(t)\;|m\rangle$$

So changing dummy variables to prepare for what follows:

$$\delta H(t)\,|\psi(t)\rangle = \sum_n c_n(t)\,\exp\left(-i\frac{E_n t}{\hbar}\right)(\delta H(t)\,|n\rangle)$$

And inserting a complete set:

$$\begin{aligned}(\delta H(t)\;|n\rangle) &= \left(\sum_m |m\rangle\langle m|\right)(\delta H(t)\,|n\rangle)\\ &= \sum_m \langle m|\delta H(t)|n\rangle\,|m\rangle\end{aligned}$$

And now putting back LHS equal to RHS

$$\sum_m i\hbar\dot{c_m}(t)\,\exp\left(-i\frac{E_m t}{\hbar}\right)|m\rangle = \sum_m\sum_n c_n(t)\,\exp\left(-i\frac{E_n t}{\hbar}\right)\langle m|\delta H(t)|n\rangle$$

Where we can now match term by term to obtain:

$$\dot{c_m}(t) = -\frac{i}{\hbar}\sum_n c_n(t)\,\exp\left(i\frac{(E_m - E_n)t}{\hbar}\right)\langle m|\delta H(t)|n\rangle \tag{1}$$

This is everything we hoped for! Notice that $H^{(0)}$ operator does not appear explicitly in Equation 1. The unperturbed Hamiltonian determines the eigenstates $\bar{m}$ and the energies E_m, but its impact on the time-dependence of the coefficients $c_m(t)$ is solely through the phase factor:

$$\exp\left(i\frac{(E_m - E_n)t}{\hbar}\right)$$

The interaction picture [1] has factored out the time dependence from the unperturbed Hamiltonian.

2 First-Order Time-Dependent Perturbation Theory

For our purposes, it suffices to consider the simple time-dependent interaction $\delta H(t)$ shown in Fig. 1. The interaction potential V has no explicit time dependence, but is turned on at t=0 and off again

[1]Formally, in the interaction picture the operators evolve as

$$O(t) = \exp(iH^{(0)}t/\hbar)\;O\;\exp(-iH^{(0)}t/\hbar)$$

while the states carry the remaining time dependence. Our derivation here achieves exactly the same separation of time dependence without invoking time evolution of the operators.

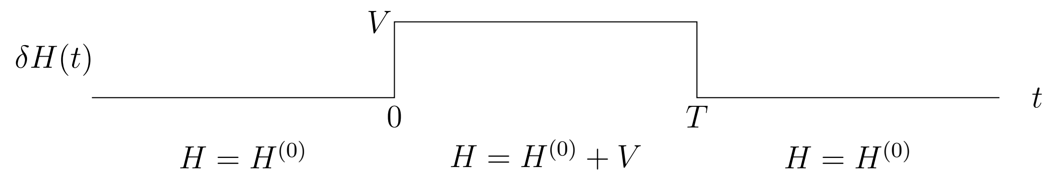


Figure 1: The interaction potential V is time-independent, however, it turns on at $t = 0$ and then off at $t = T$, making the total Hamiltonian time-dependent.

at $t = T$, making the total Hamiltonian time dependent. We will also assume that the initial state is an energy eigenstate $|i\rangle$ with

$$H^{(0)} |i\rangle = E_i |i\rangle$$

For $t \leq 0$, the interaction is absent, and $|i\rangle$ is a stationary state of total Hamiltonian, leading to the familiar time dependence:

$$|\Psi(t)\rangle = \exp\left(-i\frac{E_i t}{\hbar}\right) |i\rangle \qquad\qquad (t \leq 0)$$

This implies that:

$$c_m(0) = \delta_{mi}$$

However, starting at $t = 0$, the interaction turns on and $|i\rangle$ is no longer a stationary state. The probability p of observing the state $|m\rangle$ is given by:

$$p(t) = |c_m(t)|^2 \qquad\qquad (0 \leq t \leq T)$$

which we can no longer assume is zero for $m \neq i$, and is time dependent. At $t = T$, the interaction turns off. The states $|m\rangle$ return to being stationary states of the total Hamiltonian, but the state is no longer $|i\rangle$. We now have a probability

$$P_{fi}(T) = |c_f(T)|^2 \qquad\qquad (t \geq T)$$

of observing the system in state $|f\rangle$. We call this, the transition probability from state $|i\rangle$ to state $|f\rangle$. Notice that is is constant in time, but it does depend on how long the interaction is left on (T). The interaction potential shakes things up, mixing up the energy eigenstates, but once it turns off, things stop moving and stay in place.

Let's further assume that the interaction potential V is small compared to the time-independent Hamiltonian $H^{(0)}$.

This scenario lends itself to perturbation theory, where the new interaction $\delta H(t)$ is referred to as the perturbation.

In the interaction picture, the well-understood Hamiltonian $H^{(0)}$ is generally large compared to the time-dependent additional interaction $\delta H(t)$. Starting from the state with no additional interaction (the unperturbed state) we calculate the effect of the perturbation as a series of corrections. Because our perturbation is time-dependent, we will need time-dependent perturbation theory. This topic is covered in PHY 115B, so we will not go into the topic too deeply. Fortunately, we can obtain the first-order results in a simple, intuitive manner. In the RHS of Equation 1, we make the approximation:

$$c_n(t) \sim c_n(0) = \delta_{ni}$$

to obtain a first-order estimate:

$$\dot{c_m}(t) = -\frac{i}{\hbar} \exp\left(i\frac{(E_m - E_i)t}{\hbar}\right) \langle m|\delta H(t)|i\rangle \tag{2}$$

For any final state $|f\rangle$, we can estimate the corresponding coefficient for $t \geq T$ by an integral:

$$c_f(T) = -\frac{i}{\hbar} V_{fi} \int_0^T \exp\left(i\omega_{fi} t\right) \, dt$$

where we define the interaction potential matrix element:

$$V_{mn} = \langle m|V|n\rangle \tag{3}$$

and the angular frequency:

$$\omega_{mn} = \frac{E_m - E_n}{\hbar} \tag{4}$$

There is one subtlety, however. This prescription is only valid for $f \neq i$. At first order

$$c_i(t) = 1.$$

Although probability must flow out of the initial state as other coefficients become nonzero, probabilities are quadratic in the amplitudes, and $c_i(t)$ only changes at second order.

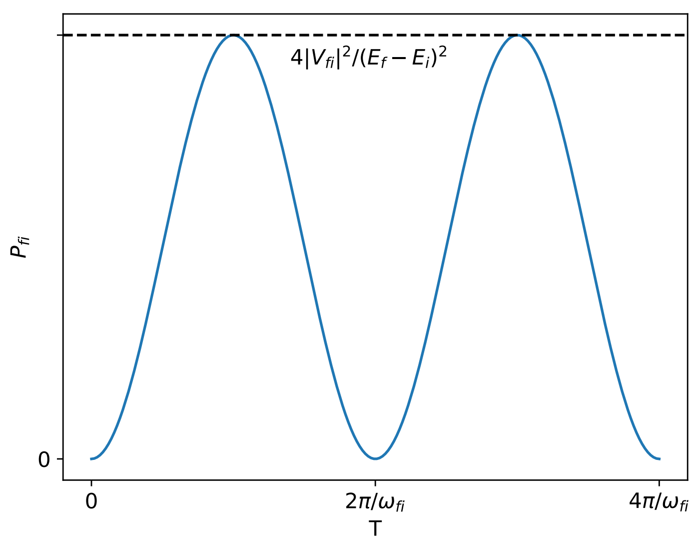


Figure 2: For $E_f \neq E_i$, the probability P_{fi} oscillates with an amplitude that goes as $1/(E_f - E_i)^2$.

Noting that:

$$\int_0^T e^{i\omega t} \, dt = e^{i\omega T/2} \frac{\sin(\omega T/2)}{\omega/2}$$

we determine the first-order transition probability (for $f \neq i$) as:

$$P_{fi}(T) = |c_f(T)|^2 = \frac{|V_{fi}|^2}{\hbar^2} \left(\frac{\sin(\omega_{fi} T/2)}{\omega_{fi}/2}\right)^2 \tag{5}$$

which is plotted in Fig. 2. Notice that T appears in the numerator, but not the denominator, of the factor:

$$F_T(\omega_{fi}) = \left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2 \tag{6}$$

In the next section, we shall see that this factor saves the day for us, twice.

3 Fermi's Golden Rule

In the previous section, we calculated the first-order transition probability $P_{fi}(T)$, for the probability of a transition from state $|i\rangle$ to $|f\rangle$, when an interaction was left on for time T. In this section, we will calculate the **rate** at which transitions occur from the initial state $|i\rangle$ to a collection of final states in the vicinity of $|f\rangle$ which we shall denote as Δf. Conceptually, that rate is:

$$\Gamma_{fi} = \frac{\sum_{\Delta f} P_{fi}(T)}{T}$$

We have no knowledge about the correct value of T. If our result is going to have any utility, Γ_{fi} must be independent of T, which can only happen if our first wish is granted:

$$\sum_{\Delta f} P_{fi}(T) \;\propto\; T \tag{7}$$

We shall see if we are so lucky!

For a first step, we'll move to the continuum by the replacement:

$$\sum_{\Delta f} \to \int_{\Delta f} dn = \int_{\Delta f} \rho(E_f)\, dE_f$$

where the integral is now over a continuous range of final states in the vicinity of $|f\rangle$, and

$$\rho(E_f) = \left|\frac{dn}{dE_f}\right| \tag{8}$$

is the Jacobian of the transformation from dn to dE. More intuitively, $\rho(E_f)$ is the number of states dn per energy range dE. We refer to $\rho(E_f)$ as the density of states. An example calculation for the non-relativistic case, using box regulation, is in the Appendix.

Using the main result from the previous section (Equation 5) we now have:

$$\sum_{\Delta f} P_{fi}(T) = \frac{1}{\hbar^2}\int_{\Delta f} \rho(E_f)\, |V_{fi}|^2 \, \left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2 \, dE_f$$

At this point, a reasonable human being would consider giving up. We don't know anything specific about the $\rho(E_f)$ and $|V_{fi}|^2$ that appear in the integral, nor do we know much about the range of integration, which we've labeled as Δf, to indicate a continuum in the vicinity of $|f\rangle$. The only way we can hope to proceed is if there is a delta function lurking in the integrand, which we shall take as our second wish.

There is precisely one factor in the integrand which we know explicitly:

$$F_T(\omega_{fi}) = \left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2$$

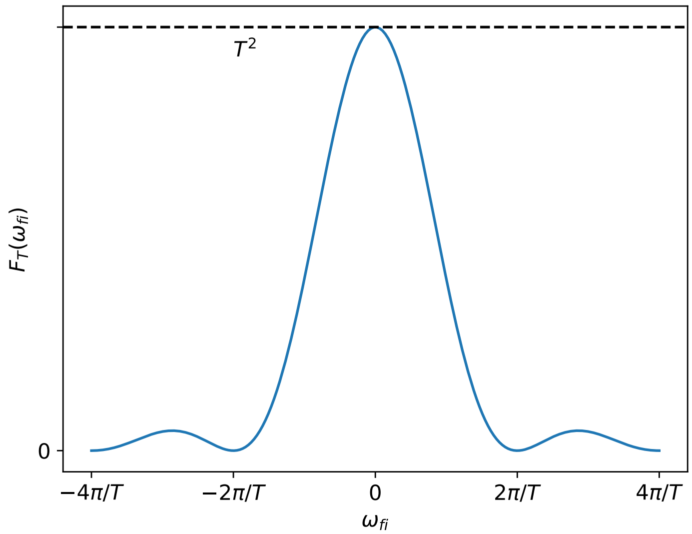


Figure 3: For large enough T, $f(\omega_{fi})$ becomes a delta function and the area under the curve is $\propto T$

and it is plotted in Fig. 3. Remarkably, it grants both of our wishes. It has amplitude T^2 and width $\propto 1/T$, so the area is portional to T, exactly as we hoped. Furthermore, as T increases, the amplitude becomes larger and the width narrower, so that it approaches a δ-function, exactly as we hoped. The delta function sets ω_{fi} to zero, which requires $E_f = E_i$.

In light of these developments, we are justified to rewrite our integral as:

$$\sum_{\Delta f} P_{fi} = \frac{1}{\hbar^2}\rho(E_i)\,|V_{fi}|^2 \int_{-\infty}^{+\infty} \left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2 \, dE_f$$

where we have taken $E_f = E_i$ and pulled out the constant factors $\rho(E_f) = \rho(E_i)$ and $|V_{fi}|^2$.. We have also extended the range of integration of E_f to $(-\infty, +\infty)$, which is justified because the integral is zero outside of a very narrow range. The remaining integral is calculable, and yields:

$$\sum_{\Delta f} P_{fi} = \frac{2\pi}{\hbar}\rho(E_i)\,|V_{fi}|^2\, T$$

and recalling:

$$\Gamma_{fi} = \frac{\sum_{\Delta f} P_{fi}}{T}$$

we have:

$$\Gamma_{fi} \;=\; \frac{2\pi}{\hbar}\,\rho(E_i)\,|V_{fi}|^2 \tag{9}$$

This result is known as Fermi's Golden Rule (FGR), and it is the foundation of particle physics.

We calculated this result with first-order time-dependent perturbation theory, so V_{fi} is just the first term in a series that turns out to be:

$$T_{fi} = \langle f|V|i\rangle + \sum_{j\neq i} \frac{\langle f|V|j\rangle\,\langle j|V|i\rangle}{E_i - E_j} + \ldots \tag{10}$$

Further noting that

$$\rho(E_i) = \left|\frac{dn}{dE_f}\right|\Bigg|_{E_i} = \int \left|\frac{dn}{dE_f}\right| \delta(E_f - E_i)dE_f = \int \delta(E_f - E_i)dn$$

we can write FGR as:

$$\Gamma_{fi} \;=\; \frac{2\pi}{\hbar} \int |T_{fi}|^2 \; \delta(E_f - E_i) \; dn \tag{11}$$

The rate of a process is dependent on the transition matrix element squared ($|T_{fi}|^2$) which describes the dynamics of the interaction. It also depends on the phase space available, consistent with conservation of energy: $\delta(E_f - E_i)\; dn$. In the next section, will clarify the phase space integral.

4 Lorentz Invariant Phase Space

We quantify the phase-space available to a single particle by integrating over it's available four-momentum:

$$d^4p \;=\; dp^0\, dp^1\, dp^2\, dp^3 \;=\; dE\, dp_x\, dp_y\, dp_z$$

First, we'll show that this is Lorentz invariant, using our knowledge of the Lorentz Group. Recall that the momentum transforms as

$$p^{\mu'} = \Lambda^{\mu'}_{\nu} p^{\nu}$$

where the matrix Λ with components $\Lambda^{\mu'}_{\nu}$ is a member of the Lorentz group. The Jacobian of a linear transformation is just the norm of the determinant of the matrix representing the linear transformation. So:

$$d^4p' \;=\; |\det\Lambda|\; d^4p$$

Now, the defining property of Lorentz transformations is:

$$\Lambda^T G \Lambda = G$$

where G is the metric tensor. Using properties of the determinant:

$$\begin{aligned}
\det\left(\Lambda^T G \Lambda\right) &= \det G \\
\det(\Lambda)\det(G)\det(\Lambda) &= \det G \\
(\det(\Lambda))^2 \det(G) &= \det G
\end{aligned}$$

from which it follows that:

$$\det\Lambda = \pm 1$$

Therefore we have:

$$(d^4p)' = |\det\Lambda|\, d^4p = d^4p$$

and the four-momentum volume is Lorentz Invariant.

An integral over d^4p will included many different particle masses via:

$$m^2 = p^\mu p_\mu = (p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2$$

We can restrict our integration to include the particle mass by including a delta function:

$$\delta(p^\mu p_\mu - m^2)$$

since $p^\mu p_\mu$ is Lorentz invariant. Howerver, this still allows $E = p^0 c$ to run over negative values, so we also include a θ-function as well:

$$\theta(p^0)$$

Note that $\theta(p^0)$ is not invariant with respect to time reversal, but it is invariant wrt to rotations (which preserve time) and boosts (which are linear). Thus we define the single particle phase-space integral:

$$d\Phi \;\equiv\; \theta(p^0)\;(2\pi)\,\delta(p^\mu p_\mu - m^2)\;\frac{d^4p}{(2\pi)^4}$$

where we have included factors of 2π consistent with our convention for the Fourier transform. Thus defined $d\Phi$ is invariant with respect to the **orthochronous** Lorentz Group which does not include time-reversal (T). This form of $d\Phi$ makes it's Lorentz Invariance manifest. However, there is another way to write it, more convenient for calculations, by noting:

$$\delta(p^\mu p_\mu - m^2) = \delta((p^0)^2 - |\vec{p}|^2 - m^2) = \delta((p^0)^2 - (E)^2)$$

and using this to complete the integral over dp^0. Note the delta function identity (proven below) that:

$$\delta(x^2 - a^2) = \frac{1}{2|a|}\left(\delta(x-a) + \delta(x+a)\right)$$

In this case, we have $p^0 > 0$ so only the first δ function matters, and we have:

$$d\Phi = \frac{d^3p}{(2E)\;(2\pi)^3}$$

When working with integrals of phase space, we have to be somewhat careful, and recall that E is an integration variable that has been set (via the delta function) to the value $E = \sqrt{m^2 + |\vec{p}|^2}$ which depends on the other integration values!

In the exercises, you will show that:

$$\delta^4(\Lambda p) = \delta^4(p) \tag{12}$$

where Λ is a Lorentz transformation. That is, the four-dimensional δ-function is Lorentz invariant.

Exercise: Verify Equation 12 using the invariance of d^4p. Hint: show that:

$$\int \delta(\Lambda p) f(p) d^4p = \int \delta(p) f(p) d^4p$$

for an arbitrary function $f(p)$, taking care to note that p is a dummy variable.

5 Normalization of Momentum Continuum States

With our single particle Lorentz invariant phase space $d\Phi$ defined, we next want to find a normalization for the momentum continuum states:

$$\langle p|p'\rangle \;=\; N(p)\;\delta^3(p - p')$$

consistent with the completeness relation written in Lorentz invariant form:

$$1 \;=\; \int d\Phi\;|p\rangle\,\langle p| \tag{13}$$

To do so, we calculate:

$$\begin{aligned}
|p'\rangle &= \int d\Phi \; |p\rangle \, \langle p|p'\rangle \\
&= \int \frac{d^3p}{(2E)(2\pi)^3} \, N(p)\delta(p-p') \; |p\rangle \\
&= \frac{N(p')}{(2E')(2\pi)^3} \; |p'\rangle
\end{aligned}$$

from which we conclude:

$$N(p) = (2E)\,(2\pi)^3$$

and so

$$\langle p|p'\rangle \;=\; (2E)\,(2\pi)^3\;\delta^3(p-p') \tag{14}$$

In the exercises, you will show that these results are consistent with making the substitution:

$$|p\rangle \rightarrow \frac{|p\rangle}{\sqrt{2E}} \tag{15}$$

into the corresponding non-relativistic equations.

Exercise: Starting from the completeness equation from NR QM:

$$1 = \int \frac{d^3p}{(2\pi)^3} \, |p\rangle \, \langle p| \, ,$$

apply the substitution of Equation 15, read off the phase space integral consistent with the new normalization, and compare to Lorentz Invariant Phase Space. □

6 Lorentz Covariant Form of Fermi's Golden Rule

In this section, we'll apply Fermi's Golden Rule to the process

$$a + b + \cdots \rightarrow 1 + 2 + \ldots$$

so that the initial state is:

$$|i\rangle = |p_a \, p_b \, \ldots\rangle = |p_a\rangle \times |p_b\rangle \times \ldots$$

and the final states is:

$$|f\rangle = |p_1 \, p_1 \, \ldots\rangle = |p_1\rangle \times |p_1\rangle \times \ldots$$

We'll use a simple notation where

$$\sum E_a = E_a + E_b + \ldots$$

and

$$\sum E_i = E_1 + E_2 + \ldots$$

and similarly for products. In this context the phase-space integral in Fermi's Golden rule becomes:

$$(2\pi)\delta(E_f - E_i)dn \rightarrow (2\pi)^4\delta^4(\sum p_i - \sum p_a) \prod_i \frac{d^3p_i}{(2\pi)^3} \tag{16}$$

where we have included all particles in the final state, but added a delta function to ensure that three-momentum is conserved. Note that no factors of $2E$ appear, because this is so far a non-relativistic formulation.

We derived Fermi's Golden Rule by considering a time interval T in some particular frame. Fermi's Golden Rule calculates rates. We do not expect rates to be invariant under special relativity, and they are not. However, we do insist that our results are consistent with special relativity. Equations need not be relativistically invariant in order to be consistent with special relativity. For example, four-momentum conservation, e.g.:

$$p_a^\mu + p_b^\mu = p_1^\mu + p_2^\mu$$

is consistent with special relativity because it can be transformed consistently (e.g. by multiplying both sides by $\Lambda_\mu^{\nu'}$ to obtain:

$$p_a^{\nu'} + p_b^{\nu'} = p_1^{\nu'} + p_2^{\nu'}$$

The values of the four-vectors are frame dependent, but the relationship between them is not. The equation

$$p^0 = m$$

is not a covariant equation. While it might be true in some particular frame, as m is a scalar and p^0 transform as the temporal part of a four vector, it cannot be true in every frame.

As we derived it, Fermi's Golden Rule is not covariant, but we can make it so. Noting that:

$$T_{fi} = \langle p_a\, p_b\, \ldots |T| p_1\, p_2\, \ldots \rangle$$

we can apply the substitution of Equation 15 to obtain the combined substitution:

$$|T_{fi}|^2 \rightarrow \frac{|M_{fi}|^2/\hbar^3}{\left(\prod 2\,E_a/\hbar^3\right)\left(\prod 2\,E_i\right)} \tag{17}$$

where M_{fi} is the new Lorentz-invariant matrix element that will replace T_{fi} in our calculations, and the factors of $\hbar$ are worked out in the exercises of Section 8. Applying Equations 16 and 17 to Fermi's Golden Rule as written in Equation 11, to obtain:

$$\Gamma_{fi} \;=\; \frac{1}{\hbar^4 \prod (2E_a/\hbar^3)} \int |\mathcal{M}_{fi}|^2\, (2\pi)^4 \delta^4\left(\sum p_a - \sum p_1\right) \prod_i \frac{d^3 p_i}{(2E_i)\,(2\pi)^3} \tag{18}$$

The integral is manifestly Lorentz invariant (recall that the four-dimensional δ-function is invariant as in Equation 12) so this form of FGR is covariant if Γ_{fi} transforms as:

$$\frac{1}{\prod E_a}$$

which, we shall see, is precisely the case.

7 Two-Body Decays

We now consider the case of a particle decaying to two particles:

$$a \rightarrow 1 + 2$$

in which FGR becomes:

$$\Gamma_{fi} = \frac{1}{\hbar 2E_a} \int |\mathcal{M}_{fi}|^2 \, (2\pi)^4 \delta^4(p_1 + p_2 - p_a) \, \frac{d^3p_1}{(2E_1)\,(2\pi)^3} \frac{d^3p_2}{(2E_2)\,(2\pi)^3} \tag{19}$$

where, as always, we take care that E_1 and E_2 are functions of the integration variables. The integral remains Lorentz invariant, so the equation is covariant as long as Γ_{fi} transforms as $1/E_a$. But Γ_{fi} is a rate, so transforms as one over the time-component of a four vector, exactly as $1/E_a$. Indeed, for this case at least, we see that FGR is a Lorentz-covariant equation.

Thus convinced of its covariance, we proceed to calculate $\Gamma_f i$ in the rest frame of the particle, where:

$$E_a = m_a, \quad \text{and} \quad \vec{p}_a = 0,$$

so that:

$$\delta^4(p_1 + p_2 - p_a) = \delta(E_1 + E_2 - m_a)\, \delta^3(\vec{p}_1 + \vec{p}_2).$$

The three-dimensional delta functions kills the integral over d^3p_2, setting:

$$\vec{p}_2 = -\vec{p}_1$$

so that:

$$\Gamma_{fi} = \frac{1}{32\pi^2 m_a \hbar} \int |\mathcal{M}_{fi}|^2 \, \delta(E_1 + E_2 - m_a) \, \frac{d^3p_1}{E_1 E_2}$$

Defining:

$$p \equiv |\vec{p}_1|$$

and changing to spherical coordinates:

$$d^3p_1 = p^2 dp d\Omega$$

we have

$$\Gamma_{fi} = \frac{1}{32\pi^2 m_a \hbar} \int |\mathcal{M}_{fi}|^2 \, \delta(u - m_a) \, \frac{p^2 dp d\Omega}{E_1 E_2}$$

where $d\Omega = d\phi d\cos\theta$ is the solid angle and:

$$u \equiv E_1 + E_2$$

Noting that:

$$\frac{du}{dp} = \frac{E_1 + E_2}{E_1 E_2} p = \frac{up}{E_1 E_2}$$

and so:

$$\frac{du}{u} = \frac{p\, dp}{E_1 E_2}$$

We now have:

$$\Gamma_{fi} = \frac{1}{32\pi^2 m_a \hbar} \int |\mathcal{M}_{fi}|^2 \, \delta(u - m_a) \, \frac{p}{u} \, du d\Omega$$

The last delta function kills the integral over u setting:

$$u = E_1 + E_2 = m_a$$

and so:

$$\Gamma_{fi} = \frac{p^*}{32\pi^2 m_a^2 \hbar} \int |\mathcal{M}_{fi}|^2 d\Omega$$

where:

$$p^* = |\vec{p}_1|$$

calculated for

$$E_1 + E_2 = m_a \quad \text{and} \quad \vec{p}_2 = -\vec{p}_1.$$

Notice that the δ-functions simply impose four-momentum conservation. It is left as an exercise to show that:

$$p^* = \frac{\sqrt{m_a^4 + m_1^4 + m_2^4 - 2(m_a^2 m_1^2 + m_1^2 m_2^2 + m_2^2 m_a^2)}}{2m_a} \tag{20}$$

Exercise: Verify Equation 20.

The decay rate Γ_{fi} that we have calculated is independent of time, as it must be, to be logically consistent with indentical indistinguishable particles, which cannot, therefore, have any "age". As a result, we have:

$$\frac{dN}{dt} = -\Gamma N$$

with solution:

$$N(t) = N(0)\exp(-\Gamma t) = N(0)\exp(-t/\tau)$$

where

$$\tau = \frac{1}{\Gamma}$$

is the lifetime of the particle. The particle lifetime is easily observable in experiments, for example, by looking at the distance particles with a known energy travel inside a particle detector before decaying (accounting for time dilation).

When a particle has multiple decay modes, the total rate is the sum of all of them:

$$\Gamma_{\text{tot}} = \sum \Gamma_i$$

and the particle lifetime is:

$$\tau = \frac{1}{\Gamma_{\text{tot}}}$$

Experiments can also measure the branching fraction of a particular decay: the ratio of the number of observed decays to a particular channel to the total number of decays:

$$Br(a \to x + y) = \frac{N(a \to x + y)}{N(a \to \text{anything})} = \frac{\Gamma(a \to x + y)}{\Gamma_{\text{tot}}}$$

8 Volume Normalization

Before proceeding, we will need to face an uncomfortable reality. If we are going to consider interacting particles, we'll need to consider the volume that our particles occupy, because the rate at which they interact will clearly depend on this. But volume is not Lorentz invariant. It turns out, one can just set the volume $V = 1$ and everything works out just fine, and that's the approach that Thomson takes. Griffiths evades the topic entirely by just giving FGR for decays and cross-sections separately without any attempt at a derivation. We are going to show how V cancels beautifully when explicitly accounted for, thus justifying taking Thomson's approach and simply setting $V = 1$ for calculations.

To normalize to one particle per volume V we put:

$$|\Psi\rangle \to |\Psi\rangle / \sqrt{V}$$

Then we need:

$$d\Phi \to V d\Phi$$

which is no longer Lorentz Invariant. However:

$$\int d\Phi \, |p\rangle \langle p| \to \int V d\Phi \, (|p\rangle / \sqrt{V})(\langle p| / \sqrt{V}) \; = \; \int d\Phi \, |p\rangle \langle p| = 1$$

so when combined with the normalized wave function, the calculations (at least in this example) are still Lorentz Invariant.

Our calculation of FGR is also modified. As shown in the Appendix, the appropriate factor is $V/\hbar^3$, and so:

$$\sum_{\Delta f} \to \int_{\Delta f} dn = V \int_{\Delta f} \rho(E_f) \, dE_f$$

So our prescription for modifying the NR FGR becomes:

$$|T_{fi}|^2 \to \frac{(V/\hbar^3)|M_{fi}|^2}{(\prod 2(V/\hbar^3)E_a)(\prod 2(V/\hbar^3)E_i)} \tag{21}$$

and

$$(2\pi)\delta(E_f - E_i)dn \to (2\pi)^4 \delta^4(\sum p_i - \sum p_a) \prod_i \frac{(V/\hbar^3)d^3p_i}{(2\pi)^3} \tag{22}$$

It is left as exercise to show that the volume normalization, when applied consistently, has no impact on the decay rate calculation. In the next section, we will show that for scattering, an additional volume factor in FGR cancels a volume factor in the flux.

Exercise: Set $V = 1$ in Equations 21 and 22 and apply the substitution to Fermi's Golden Rule as written in Equation 11. Show that you obtain Equation 18 including the factors of $\hbar$.

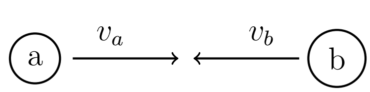
Exercise: Show that Equation 19 is unchanged when using the substitutions in Equations 21 and 22, which explicitly including the volume factors,instead of Equations 16 and 17.

9 Interaction Cross-Section

Next we will consider a scattering process:

$$a + b \to 1 + 2 + 3 + \ldots$$

Eventually, we will arrive at Lorentz Invariant results, but we'll orient ourselves first in a frame where the relative velocities of a and b are collinear, as in:



Suppose that in a volume V, we have N_a particles of type a, each with velocity v_a, and N_b particles of type b, each with velocity v_b located in a volume V, with velocities all collinear as in the diagram. We would expect the rate to scale as:

$$\Gamma \propto (N_a/V)\,(N_b/V)\,V = N_a N_b / V$$

Now suppose further that the *dynamics* of the interaction causing a and b to scatter does not depend on their relative velocity. The rate of scattering will scale linear with the relative velocity

$$\Gamma \propto v_a + v_b$$

because a and b particles will encounter one another more frequently. Putting these effects together we have:

$$\Gamma \propto N_a N_b \frac{(v_a + v_b)}{V} = \frac{(v_a + v_b)}{V}$$

where in the last step we have set $N_a = 1$ and $N_b = 1$ to consider a single pair of interacting particles. We define the luminosity

$$L = \frac{v_a + v_b}{V}. \tag{23}$$

which contains the factors that influence the rate independently of the dynamics of the interaction. We define the interaction cross section (σ) as the proportionality constant between the luminosity (L) and the rate (Γ):

$$\Gamma = L\,\sigma \tag{24}$$

Note that σ must have units of area because

$$\Gamma[1/t] = L[1/At]\;\sigma[A]$$

In the exercises, you will show that for elastic scattering of a point particle on a solid object, the interaction cross-section is quite literally the cross-sectional area of the solid object. More generally, the cross-section is an accounting trick: the rate of a scattering process is precisely the rate at which particles of type a and b are brought to within a transverse area σ of one another.

Fermi's Golden Rule determines the rate gamma for the scattering process

$$a + b \to 1 + 2 + 3 + \ldots$$

as:

$$\Gamma_{fi} \;=\; \frac{1}{V}\,\frac{\hbar^2}{2E_a 2E_b} \int |\mathcal{M}_{fi}|^2\;(2\pi)^4\delta^4\big(\sum p_a - \sum p_1\big) \prod_i \frac{d^3p_i}{(2E_i)\,(2\pi)^3}$$

where we are (for now) keeping the volume factors (V) explicit, by using the substitutions in Equations 21 and 22. Unlike the case for the two-body decay of Equation 19, the volume factors V do not all cancel at this stage. The additional initial state particle brings in one more factor of V which is not cancelled. However, using Equations 23 and 24 we have:

$$\sigma \;=\; \frac{\hbar^2}{4E_aE_b(v_a + v_b)} \int |\mathcal{M}_{fi}|^2\;(2\pi)^4\delta^4\big(\sum p_a - \sum p_1\big) \prod_i \frac{d^3p_i}{(2E_i)\,(2\pi)^3}$$

As before, the integral is manifestly Lorentz invariant, so σ will transform as $1/F$ where:

$$F = 4E_aE_b(v_a + v_b)$$

In the exercise below, you will show that F can be written in the equivalent form (for this particular frame):

$$F = 4\sqrt{(p_a \cdot p_b)^2 - (m_a m_b)^2} \tag{25}$$

which is manifestly Lorentz invariant.

Exercise: Calculate the Lorentz invariant phase-space:

$$F = 4\sqrt{(p_a \cdot p_b)^2 - (m_a m_b)^2}$$

in a frame where the velocities of a and b are collinear, and show that:

$$F = 4E_a E_b(v_a + v_b) = 4(E_a p_b + E_b p_a) \quad \square$$

We can therefore write the cross-section as:

$$\sigma = \frac{\hbar^2}{4\sqrt{(p_a \cdot p_b)^2 - (m_a m_b)^2}} \int |\mathcal{M}_{fi}|^2 \, (2\pi)^4 \delta^4(\sum p_a - \sum p_1) \prod_i \frac{d^3 p_i}{(2E_i)\,(2\pi)^3}$$

which is manifestly Lorentz Invariant. This is excellent news, as the interaction cross-section encodes the dynamics of the interactions between particles a and b, and therefore must be Lorentz invariant as the laws of physics must be the same in all inertial frames.

The interaction cross-section is bread-and-butter experimental and theoretical particle physics. It is an excellent experimental handle, applicable to the center of a star to the surface of our ocean. In the simplist form, we aim particles of type a at particles of type b and count the number that scatter during the course of our experiment.

In the exercise below, you will show that in the CMS, the golden rule for the two-body scattering cross-section is:

$$\sigma = \frac{\hbar^2 p_f^*}{64\pi^2 s p_i^*} \int |\mathcal{M}_{fi}|^2 d\Omega \tag{26}$$

where $s = (p_a + p_b)^2$ is the total energy in the CMS. Hint: show that the
Exercise: Show that in the CMS, the Lorentz invariant flux is

$$F = 4\,\sqrt{s}\,p_i^*$$

A Density of Final States

We cannot count continuum states, so we impose a fiction that a free particle is in fact constrained to a box with sides of length L and so:

$$V = L^3$$

We hope that our final results will not depend on this fictitious volume V. With this in mind, we consider the normalization of the energy eigenstates of the free particle (plane waves) as:

$$\psi(x) = N e^{ik_x x}\, e^{ik_y y}\, e^{ik_z z}$$

If we choose:

$$N = \sqrt{\frac{1}{V}}$$

then:

$$\int_V |\psi(x)|^2 d^3x = 1$$

Assuming periodic boundary conditions:

$$\begin{aligned} k_x L &= 2\pi n_x \\ k_y L &= 2\pi n_y \\ k_z L &= 2\pi n_z \end{aligned}$$

so that

$$dn = dn_x dn_y dn_z = \left(\frac{L}{2\pi}\right)^3 dk_x dk_y dk_z$$

and we have:

$$dn = \left(\frac{L}{2\pi}\right)^3 d^3k \tag{27}$$

in terms of the momentum $p = \hbar k$, we have:

$$dn = \frac{V/\hbar^3}{(2\pi)^3}\, d^3p \tag{28}$$

which reproduces the single-particle phase-space factors in Equation 22, including the factors of $\hbar$. Note that in the convention here, dn is dimensionless, consistent with our framing of the problem as counting discrete states of a particle constrained to a finite volume V.

It is convenient to express Equation 27 as a differential in energy. As our box size cannot possibly be Lorentz invariant, we'll use the non-relativistic energy of the free particle (for now):

$$E = \frac{(\hbar k)^2}{2m}$$

so we have:

$$\begin{aligned}
dk &= \frac{m}{\hbar^2 k} \\
d^3k &= k^2 dk d\Omega = \frac{mk}{\hbar^2} d\Omega dE \\
dn &= \left(\frac{L}{2\pi}\right)^3 \left(\frac{m}{\hbar^2 k}\right) d\Omega\, dE
\end{aligned}$$

where we have identified the (classical) density of states:

$$\rho(E) \equiv \left|\frac{dn}{dE}\right| = \left(\frac{L}{2\pi}\right)^3 \left(\frac{m}{\hbar^2 k}\right) d\Omega \tag{29}$$

with:

$$dn = \rho(E)\, dE \tag{30}$$

B Math Tools

In the previous section, we used some math tools. First, we need to know how to calculate the Jacobian of a coordinate transformation. Suppose we are calculated an integral:

$$I = \int f(u, w)\, du\, dw$$

but suppose we have some other coordinates r and s with:

$$\begin{aligned}
u &= u(r, s) \\
w &= w(r, s)
\end{aligned}$$

then we can calculate our integral in terms of coordinates r and s via:

$$I = \int f(u(r,s), w(r,s)) \left| \frac{\partial(u,w)}{\partial(r,s)} \right| \, dr\, ds$$

where

$$\frac{\partial(u,w)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial w}{\partial r} & \frac{\partial w}{\partial s} \end{vmatrix}$$

is the Jacobian: the determinant of the matrix of partial derivatives. We can summarize this as:

$$du\, dw = \left| \frac{\partial(u,w)}{\partial(r,s)} \right| \, dr\, ds$$

The result easily extends to more coordinates, e.g.:

$$du\, dv\, dw = \left| \frac{\partial(u,v,w)}{\partial(r,s,t)} \right| \, dr\, ds\, dt$$

where now the Jacobian will be the determinant of 3x3 matrix.

Second, we needed:

$$\delta(x^2 - a^2) = \frac{1}{2|a|} \left(\delta(x-a) + \delta(x+a)\right)$$

There's a trick for finding identities like this, using the Heaviside step function:

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$

and noting that

$$\int_{-\infty}^{x} \delta(y) dy = \theta(x)$$

so we can consider the delta function to be the derivative of the Heaviside step function:

$$\delta(x) = \theta'(x)$$

To find our identity, we work out that:

$$\theta(x^2 - a^2) = \theta(x - |a|) + (1 - \theta(x + |a|))$$

Take the derivative of both sides:

$$\begin{aligned} \theta'(x^2 - a^2) 2x &= \theta'(x - |a|) - \theta'(x + |a|) \\ \theta'(x^2 - a^2) &= \frac{\theta'(x - |a|)}{2x} - \frac{\theta'(x + |a|)}{2x} \end{aligned}$$

Now identify $\delta(x)$ with $\theta'(x)$ and simplifying:

$$\begin{aligned} \delta(x^2 - a^2) &= \frac{\delta(x - |a|)}{2x} - \frac{\delta(x + |a|)}{2x} \\ &= \frac{\delta(x - |a|)}{2|a|} - \frac{\delta(x + |a|)}{-2|a|} \\ &= \frac{\delta(x - |a|) + \delta(x + |a|)}{2|a|} \\ &= \frac{\delta(x - a) + \delta(x + a)}{2|a|} \end{aligned}$$